V-BOND CL400

Revision: 02

Revision date 29/09/2023

SECTION 1: Identification of the substance/mixture and of the company/undertaking		
1.1	**Product identifier**	V-BOND CL400
1.2	**Relevant identified uses of the substance or mixture and uses advised against**	
	Relevant identified uses:	Coating paper
1.3	**Details of the supplier of the safety data sheet**	
	Company:	VIV INTERCHEM 22 Sukhumvit 42, Sukhumvit Rd, Prakanong, Klongtoey, Bangkok 10110 Telephone: +662 712 1044 E-mail address: thasanaprapha@vivgroup.com
1.4	**Emergency telephone number**	
	International emergency number:	Telephone: 02-7094600-4 #214

SECTION 2: Hazards Identification		
2.1	**Classification according to UN GHS 2009**	
	Classification of the substance and mixture:	
	Acute toxicity:	Cat. 5 (oral)
	Acute toxicity:	Cat. 4 (Inhalation-mist)
	Skin corrosion/irritation:	Cat.2
	Serious eye damage/eye irritation:	Cat. 2B
	Skin sensitization:	Cat.1
	Germ cell mutagenicity:	Cat.2
	Specific target organ toxicity- single exposure:	Cat. 3 (irritating to respiratory system)
	Hazardous to the aquatic environment – acute:	Cat.3
	Label elements and precautionary statement: Pictogram: Signal word: Warning Hazard Statement:	
	H320	Causes eye irritation
	H315	Causes skin irritation

	H332	Harmful if inhaled.
	H303	May be harmful if swallowed.
	H317	May cause an allergic skin reaction
	H335	May cause respiratory irritation.
	H341	Suspected of causing genetic defects.
	H402	Harmful to aquatic life.
	Precautionary Statements (Prevention):	
	P280	Wear protective gloves/protective clothing/eye protection/ face protection.
	P271	Use only outdoors or in a well-ventilated area.
	P260	Do not breathe mist or vapour.
	P201	Obtain special instructions before use.
	P273	Avoid release to the environment.
	P202	Do not handle until all safety precautions have been read and understood.
	P272	Contaminated work clothing should not be allowed out of the workplace.
	P264	Wash with plenty of water and soap thoroughly after handling.
	Precautionary Statements (Response):	
	P308+P311 IF exposed or concerned:	Call a POISON CENTER or doctor/physician.
	P305+P351+P338 IF IN EYES:	Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
	P304 + P340 IF INHALED:	Remove person to fresh air and keep comfortable for breathing.
	P303 + P352 IF ON SKIN (or hair):	Wash with plenty of soap and water.
	P332 + P313 if skin irritation occurs:	Get medical advice/attention.
	P362 + P364	Take off contaminated clothing and wash it before reuse.
	P337 + P311 If eye irritation persists:	Call a POISON CENTER or doctor/physician.
	Precautionary Statements (Storage):	
	P403 + P233	Store in a well-ventilated place. Keep container tightly closed. Store locked up.
	Precautionary Statements (Disposal):	
	P501	Dispose of contents/container to hazardous or special waste collection point
	Other hazards which do not result in classification:	
	If applicable information is provided in this section on other hazards which do not result in Classification but which may contribute to the overall hazards of the substance or mixture.	

SECTION 3: Composition/information on ingredients		
3.1	**Chemical nature:**	Mixture , Glyoxal (content (W/W)+ N/A : 40%, Water (content (W/W): 60%
3.2	**Hazardous ingredients:**	
	Glyoxal Content (W/W): >=39.5% - <=40.5% CAS Number: 107-22-2	Acute Tox.: Cat. 5 (oral) Acute Tox.: Cat. 4 (Inhalation – mist) Skin Corr./Irrit.: Cat. 2 Eye Dam./Irrit.: Cat. 2B Skin Sens.: Cat. 1 Muta.: Cat. 2 STOT SE: Cat. 3 (irr. to respiratory syst.) Aquatic Acute: Cat. 3
	Ethyleneglycol Content (W/W): >= 0% - <= 2.5% Acute CAS Number: 107-21-1	Tox.: Cat. 4 (oral) STOT RE (Kidney): Cat. 2

SECTION 4: First aid measures		
4.1	**Different measures for:**	
	General advice:	Remove contaminated clothing.
	If inhaled:	Keep patient calm, remove to fresh air, seek medical attention.
	On skin contact:	Wash thoroughly with soap and water.
	On contact with eyes:	Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.
	On ingestion:	Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.
	Note to physician:	
	Symptoms:	The most important known symptoms and effects are described in the labelling (see section 2) and/or in section 11.
	Treatment:	Treat according to symptoms (decontamination, vital functions), no known specific Antidote.

SECTION 5: Firefighting measures		
	Suitable extinguishing media:	Water spray, foam, dry powder, carbon dioxide
	Specific hazards:	Nitogen oxides, carbon oxides The substances/groups of substances mentioned can be released in case of fire.
	Special protective equipment:	Wear self-contained breathing apparatus and chemical-protective clothing.
	Further information:	Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems.

SECTION 6: Accidental release measures		
6.1	**Personal precautions:**	Avoid contact with the skin, eyes and clothing.
6.2	**Environmental considerations:**	Do not discharge into drains/surface waters/groundwater.
6.3	**Methods for Cleaning up or taking up:**	
	For large amounts:	Pump off product.
	For residues:	Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr). Dispose of absorbed material in accordance with regulations

SECTION 7: Handling and storage		
7.1	**Handling:**	Ensure thorough ventilation of stores and work areas.
7.2	**Storage:**	
	Suitable materials for containers:	Stainless steel 1.4401, Stainless steel 1.4301(V2), High density polyethylene (HDPE), glass, Low density polyethylene (LDPE)
	Unsuitable materials for containers:	Paper/Fiberboard
	Further information on storage conditions:	Protect from air.
	Storage temperature:	< 50 ˚C
	Storage duration:	6 Months. May yellow after lengthy storage. During the storage a product characteristic clouding or crystallin precipitation of the trimeric Glyoxal hydrate may occur. The process is reversible if warmed up mild (max. 40 ˚C). From the data on storage duration in this safety data sheet no agreed statement regarding the Warrantee of application properties can be deduced.

	Protect from temperatures above:	50 ˚C, The packed product must be protected against exceeding the indicated temperature.
SECTION 8: Exposure controls/personal protection		
8.1	**Components with occupational exposure limits:**	
	Ethyleneglycol	107-21-1; CLV 100 mg/m^3 (ACGIHTLV), aerosol
	Glyoxal	107-22-2; TWA value 0.1 mg/m^3 (ACGIHTLV), inhalable fraction and vapor
8.2	**Personal Protective Equipment :**	
	Eye/Face Protection:	Safety glasses with side-shields (frame goggles) (e.g. EN166)
	Body Protection:	Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit, (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).
	Hand protection:	Suitable chemical resistant safety gloves (EN374) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN374) : E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc. Manufacturer's directions for use should be observed because of great diversity of types.
	Respiratory Protection:	Wear respiratory protection if ventilation is inadequate. Gas filter for gasses/vapours of organic compounds (boiling point < 65 ˚C f.e. EN 14387 Type AX)
	General safety and hygiene measures:	Avoid contact with the skin, eyes and clothing. When using, do not eat, drink or smoke. Avoid contact with skin and eyes. Remove soiled or soaked clothing immediately. Wash off any contamination that gets onto the skin with plenty of water and soap, skin care.

SECTION 9: Physical and chemical properties		
9.1	**Information on basic physical and chemical properties**	
	Appearance:	Liquid
	Colour:	Colourless to yellow
	Odor:	Product specific
	Odour threshold:	Not determined due to potential health hazard by inhalation.
	pH value:	2 – 3.5 (400 g/l, 20 ˚C)
	Melting range:	-50 - -15 ˚C (OECD Guideline 102)
	Boiling point:	103.6 ˚C (1.013 hPa) (Directive 92/69/EEC, A.2)
	Flash point:	Non-flammable
	Evaporation rate:	Value can be approximated from Henry's Law constant or vapour pressure.
	Flammability (solid/gas):	Not flammable

	Lower explosion limit:	For liquids not relevant for classification and labelling., The lower Explosion point may be 5 -15 ˚C below the flash point.
	Upper explosion limit:	For liquids not relevant for classification and labelling.,
	Ignition temperature:	Approx. 285 °C (DIN 51794)
	Thermal decomposition:	No decomposition if correctly stored and handled.
	Self ignition:	Based on its structural properties the product is not classified as self-igniting. Test type: Spontaneous self-ignition at room-temperature.
	Self heating ability:	It is not a substance capable of spontaneous heating.
	Vapour pressure:	20.2 hPa (20 °C) information applied to the solvent. 106.7 hPa (50 °C) information applied to the solvent. (Directive 92/69/EEC, A.4)
	Density:	Approx. 1.27 g/cm^3 (20 °C) (Directive 92/69/EEC, A.3) 1.2514 g/cm^3 (50 °C)
	Relative density:	1.27 (20 °C) (Directive 92/69/EEC, A.3)
	Solubility in water:	Miscible (approx. 20 °C)
	Partitioning coefficient n-octanol/water (log Pow):	-1.15 (approx. 23 °C; pH value: 7) (OECD Guideline 107)
	Adsorption/water-soil:	KOC: 2.1; log KOC: 0.32 (OECD Guideline 121)
	Surface tension:	Based on chemical structure, surface activity is not to be Expected.
	Viscosity, dynamic:	8.37 mPa.s (20 °C) (OECD 114) 4.25 mPa.s (40 °C)
	Viscosity, kinematic:	6.6 mm^2/s (20 °C) (OECD 114) 3.38 mm^2/s (40 °C)
	Molar mass:	58.04 g/mol

SECTION 10: Stability and reactivity		
10.1	**Conditions to avoid:**	Temperature:> 50 °C See MSDS section 7 – Handling and storage.
10.2	**Thermal decomposition:**	No decomposition if correctly stored and handled.
10.3	**Substances to avoid:**	strong alkalies
10.4	**Corrosion to metals:**	No corrosive effect on metal.
10.5	**Hazardous reactions:**	Reacts with strong alkalies. Exothermic reaction.
10.6	**Hazardous decomposition products:**	No hazardous decomposition products if stored and handled as prescribed/indicated.

SECTION 11: Toxicological information		
11.1	**Acute toxicity**	Of low toxicity after single ingestion. Virtually nontoxic after a single skin contact. Of moderate toxicity after short-term inhalation.
	Experimental/calculated data:	LD50 rat (oral): 3,300 mg/kg (OECD Guideline 401) LC50 rat (by inhalation): 2.44 mg/l 4 h (OECD Guideline 403) An aerosol was tested. LD50 rat (dermal): > 2,000 mg/kg (OECD Guideline 402) Limit concentration test only (LIMIT test).
11.2	**Irritation effects**	Eye contact causes irritation. Skin contact causes irritation.
	Experimental/calculated data:	Skin corrosion/irritation rabbit: Irritant. (OECD Guideline 404) Serious eye damage/irritation rabbit: Irritant. (OECD Guideline 405)
11.3	**Respiratory/Skin sensitization**	Caused skin sensitization in animal studies. Caused sensitization in humans.
	Experimental/calculated data:	Guinea pig maximization test guinea pig: skin sensitizing (OECD Guideline 406) Human Maximization test human: skin sensitizing (Patch test) Literature data.
11.3	**Germ cell mutagenicity**	The substance was mutagenic in various test systems with microorganisms and cell cultures; However, these results could not be confirmed in tests with mammals. Mutagenic properties can not be excluded on the basis of experimental data.
11.4	**Carcinogenicity**	The whole of the information assessable provides no indication of a carcinogenic effect.
11.5	**Reproductive toxicity**	The results of animal studies gave no indication of a fertility impairing effect.
11.6	**Developmental toxicity**	No indication of a developmental toxic / teratogenic effect were seen in animal studies.
11.7	**Specific target organ toxicity (single exposure)**	Causes temporary irritation of the respiratory tract.
11.8	**Repeated dose toxicity and Specific target organ toxicity (repeated exposure)**	After repeated exposure the prominent effect is local irritation.
11.9	**Aspiration hazard**	No aspiration hazard expected.

SECTION 12: Ecological information		
12.1	Ecotoxicity	
	Assessment of aquatic toxicity:	Acutely harmful for aquatic organisms. Based on long-term (chronic) toxicity study data. The product is very likely not harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.
	Toxicity to fish:	LC50 (96 h) > 186 - <272 mg/l, Leuciscus idus (DIN 38412 Part 15, static) The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.
	Aquatic invertebrates:	EC50 (48 h) 161 mg/l. Daphnia magna (Directive 79/831/EEC, static) Nominal concentration. The ecological data given are those of the active ingredient. LC50 (96 h) 76.0 mg/l. Americamysis bahia (other, semistatic) The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.
	Aquatic plants:	EC50 (72 h) > 40 mg/l (growth rate), Scenedesmus subspicatus (OECD Guideline 201, static) The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient. EC10 (72 h) > 10 mg/l (growth rate), Scenedesmus subspicatus (OECD Guideline 201,static) The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient. EC50 (72 h) 347.1 mg/l (growth rate), Skeletonema costalum (OECD Guideline 201) The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient. No observed effect concentration (72 h) 118.4 mg/l (growth rate), Skeletonema costatum (OECD Guideline 201) The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.
	Microorganisms/Effect on activated sludge:	EC10 (16 h) 22.8 mg/l, Pseudomonas putida (DIN 38412 Part 8, static) The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient. EC50 (0.5 h) > 400 mg/l, activated sludge (OECD Guideline 209, static) The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.
	Chronic toxicity to fish:	No observed effect concentration (34 d) 112 mg/l, Pimephales promeles (OPP 72-4 (EPAGuideline), Flow through.) The ecological data given are those of the active ingredient. The statement of the toxic effect relates to the analytically determined concentration.

	Chronic toxicity to aquatic invertebrates:	No observed effect concentration (21 d), 3.19 mg/l, Daphnia magna (OECD Guideline 211, semistatic) The statement of the toxic effect relates to the analytically determined concentration. The ecological data given are those of the active ingredient.
	Soil living organisms:	LC50 (14 d) > 135 mg/kg, Eisenia foetida (OECD Guideline 207, artificial soil) The ecological data given are those of the active ingredient. The statement of the toxic effect relates to the analytically determined concentration. EC50 (28 d) >1,054 mg/kg, soil dwelling microorganisms (OECD 217, natural soil) The details of the toxic effect relate to the nominal concentration. The data have been calculated from values for a preparation with a lower substance concentration. The ecological Data given are those of the active ingredient. EC50 (28 d) >1,054 mg/kg, soil dwelling microorganisms (OECD 216, natural soil) The details of the toxic effect relate to the nominal concentration. The data have been calculated from values for a preparation with a lower substance concentration. The ecological Data given are those of the active ingredient.
	Terrestrial plants:	No observed effect concentration (21 d), Brassica napus (OECD Guideline 208) The ecological data given are those of the active ingredient. The statement of the toxic effect relates to the analytically determined concentration.
	Other terrestrial non-mammals:	No data available.
12.2	**Mobility**	
	Assessment transport between environmental compartments:	The substance will not evaporate into the atmosphere from the waste surface. Adsorption to solid soil phase is not expected.
12.3	**Persistence and degradability**	
	Information on Stability in water (hydrolysis):	The product has not been tested. The statement has been derived from the structure of the product.
12.4	**Sum parameter**	
	Chemical oxygen demand (COD):	350 mg/g
	Biochemical oxygen demand (BOD) Incubation period 5 d:	175 mg/g
12.5	**Bioaccumulation potential**	
	Assessment bioaccumulation potential:	Significant accumulation in organisms is not to be expected.

	Bioaccumulation potential:	Bioconcentration factor: 3.2, Fish (calculated)

SECTION 13: Disposal considerations		
	Incinerate in suitable incineration plant, observing local authority regulations. A waste code in accordance with the Euopean waste catalog (EWC) cannot be specified, due to dependence on the usage. The waste code in accordance with the European waste catalog (EWC) must be specified In cooperation with disposal agency/manufacturer/authorities.	
	Contaminated packaging:	Uncontaminated packaging can be recycled.

SECTION 14: Transport information		
14.1	**Domestic transport:**	Not classified as a dangerous good under transport regulations
14.2	**Sea transport**	
	IMDG	Not classified as a dangerous good under transport regulations
14.3	**Air transport**	
	IATA/ICAO	Not classified as a dangerous good under transport regulations

SECTION 15: Regulatory information		
	Hazard determining component(s) for Labelling:	Glyoxal
	Other regulations	-

SECTION 16: Other information	
	Any other intended applications should be discussed with the manufacturer.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. The data do not describe the product's properties (product specification). Neither should any agreed property nor the suitability of the product for any specific purpose be deduced from the data contained in the safety data sheet. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

END OF DATA SHEET.